

Sadeep Ariyarathna

Systems Engineer | Mechatronics Engineer

Personal Information

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Professional Summary

Systems and Mechatronics Engineer with practical experience in E/E architecture, embedded systems, and model-based development. Experienced in designing automated workflows, KPI frameworks, and control systems using Python, C/C++, MATLAB, and Azure DevOps. Skilled at integrating mechanical, electrical, and software components to develop data-driven engineering solutions. Recently completed a Master of Science in Systems Engineering and Engineering Management at Fachhochschule Südwestfalen, with a focus automation and mechatronics systems in the German engineering sector.

Experience

Intern / Masterstudent - E/E Architecture

CLAAS E-Systems GmbH, Dissen am Teutoburger Wald, Germany

Oct 2024 - Aug 2025

- Developed and implemented an **automated CI/CD workflow** using **Azure DevOps** and **LemonTree** to integrate system architecture models more efficiently, reducing merge time by ~75%.
- Created a KPI-based evaluation framework to assess workflow performance using user feedback, time trials, and success rates.
- Contributed to improving system usability (**Enterprise Architects, Lemontree and Azure DevOps**) and process transparency within the E/E architecture development team.

Mechatronics Engineer

AtlasLabs (Pvt.) Ltd, Sri Lanka

Sep 2022 - Feb 2023

- Engineered and tested mechanical components and systems from concept to launch using **Fusion 360**, reducing product development cycle time by 20%.
- Created manufacturable designs and validated them according to **Design for Manufacturing (DfM)** and **Design for Assembly (DfA)** principles.
- Reduced troubleshooting time by 35% by introducing standardized **bills of materials and validation documentation**.

Engineering Intern

MAS Intimates (Pvt.) Ltd, Sri Lanka

Jul 2021 - Feb 2022

- Designed and developed **SolidWorks**-based attachments for sewing machines, reducing material cutting time by 65%.
- Created mechanical control systems using **Arduino** and **PLC (Delta)** to evaluate multiple conceptual prototypes.
- Supported cross-functional teams in implementing manufacturing process improvements.

Education

Master of Science in Systems Engineering and Engineering Management

Fachhochschule Südwestfalen, Soest, Germany

Oct 2022 – Sept 2025

- Thesis: *A Collaboration Workflow for Merging System Architecture Models – An Example from the Agriculture Industry (Marks - 1.0)*
- Final GPA – 1.6

Bachelor of Engineering in Mechatronics

General Sir John Kotelawala Defence University, Sri Lanka

Jan 2017 – Dec 2021

- Thesis: *Development of Elbow Prosthetic Control System*
- Final GPA – 1.8

Projects

Grid Solving with Reinforcement Learning

Implemented a custom grid environment and analyzed Q-learning, Deep Q-learning, and A2C algorithms using **PyTorch**, **OpenAI Gym**, and **Stable Baselines3**.

Chess Game with Reinforcement Learning Agent

Developed a fully functional chess game with move validation, history, and timing logic. Integrated a DQN-based **reinforcement learning** agent to train autonomous gameplay strategies.

Gesture Control System

Created a computer vision program using **OpenCV** and **Python** to recognize specific finger gestures for controlling system functions such as opening applications or typing commands.

Remote Temperature Detection and Transmission

Developed a **C**-based **STM32** application capable of measuring, storing, and transmitting temperature data via GSM (**MQTT** protocol).

Prosthetic Elbow Concept

Engineered a **C++**-based control system for a prosthetic elbow using a brushless DC motor. Designed and optimized the gear system in **SolidWorks** for precision and stability.

Technical Skills

Programming	Python, C/C++, MATLAB, Arduino, PLC (Delta)
Modeling & Design	Enterprise Architect, SolidWorks, Fusion 360, MATLAB Simulink, Autodesk Eagle, STM32 CubeIDE, KiCAD
Developer Tools	Git, Azure DevOps, Docker, Jenkins, Flask
Libraries & Frameworks	PyTorch, TensorFlow, Stable Baselines, OpenAI Gym, OpenCV
Other Tools	Microsoft Office, Adobe Photoshop, HTML & CSS

Languages

German	A2 (Conversational, Learning)
English	C1 (Fluent)

Additional Skills

- Hands-on experience with **workshop tools**, **soldering**, and **PCB design**
- Strong analytical and problem-solving mindset